Lab No: 11

✵ **Lab Title:** **Strings**

- **Learning objectives:**

The following topics will be covered in this lab session

- What is a String
- When to use strings
- Declaration of strings
- Initialization of strings
- Accessing strings element

STRINGS:

It is defined as character array that is terminated by a null. A null character is either specified by using '\0' or zero. Two kinds of strings are commonly used in C++i.e. C-strings and strings .both are objects of the string class. C-strings are arrays of type char. These strings are known as *C-strings*, or *C-style strings*, because they were the only kind of strings available in the C language.

- **Example # 1:**

Write code for a program that asks the user to enter a string, and places this string in the string variable. Then it displays the string.

```
// stringin.cpp
// simple string variable
#include <iostream>
Using namespace std;
int main()
{
const int MAX = 80;   //max characters in string
char str[MAX];           //string variable str
cout << "Enter a string: ";
cin >> str; //put string in str
//display string from str
cout << "You entered: " << str << endl;
return 0;
}
```

Output:

```
Enter a string: Malakand
You entered: Malakand
```

- **Example # 2:**

There is no built-in mechanism in C++ to keep a program from inserting array elements outside an array. However, it is possible to tell the >> operator to limit the number of characters it places in an array. This Example demonstrates this approach.

```
// safetyin.cpp
// avoids buffer overflow with cin. width
#include <iostream>
#include <iomanip> //for setw
int main()
{
const int MAX = 20; //max characters in string
char str[MAX]; //string variable str
cout << "\nEnter a string: ";
cin >> setw(MAX) >> str; //put string in str,
// no more than MAX chars
cout << "You entered: " << str << endl;
return 0;
}
```

Output:

```
Enter a string: umair
You entered: umair
```

Lab Tasks

- **Task 1: String Length Calculation:**

  1. Write a C program that declares and initializes a string variable called "message" with a phrase of your choice.
  2. Use a built-in function to calculate and display the length of the string.

➯ Input:

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string message = "Hello, World!";
    cout<<message<< " Size :" << message.length() << endl;
    return 0;
}
```

➯ Output:

```
Hello, World! Size :13
```

- **Task 2: String Concatenation**

  1. Declare two string variables called "first Name" and "last Name" and initialize them with your first and last name.
  2. Concatenate the two strings to form a full name and display it.

➯ Input;

```
#include <iostream>

#include <string>

using namespace std;

int main() {

    string firstName = "Umair";
```

```
    string lastName = "Khan";
    string fullName = firstName + " " + lastName;
    cout << "The full name is: " << fullName << endl;
    return 0;
}
```

⇘ Output:

```
The full name is: Umair  Khan
```